

### Session 3 Quiz - What is Machine Learning?

#### Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:56691 ]

Fill in the blank: Machine Learning is when computers learn patterns from \_\_\_\_\_ instead of following a fixed list of instructions.

- A. examples
- B. hardware
- C. batteries
- D. magic

Correct Answer: A. examples

Explanation: Machine learning relies on being shown many instances (examples) to identify underlying characteristics rather than being told exactly what to do. Just as you learned what a dog looks like by seeing many different dogs, AI learns from examples. Hardware, batteries, and magic are not the correct basis for machine learning.

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Q2 [Type: MCQ | Difficulty: Easy | QID:56692 ]

Fill in the blank: In the AI Recipe, the examples that the AI learns from are called \_\_\_\_\_.

- A. Predictions
- B. The Rulebook
- C. The Microwave
- D. Data

Correct Answer: D. Data

Explanation: Data consists of the photos, numbers, text, or other information that serve as the learning material for the algorithm. Data is the raw ingredient or fuel that AI needs to train. Predictions are outputs, rulebooks are for traditional programming, and microwave is unrelated.

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Q3 [Type: MCQ | Difficulty: Easy | QID:56693 ]

Fill in the blank: A system that does the exact same thing every time without changing is a \_\_\_\_\_ system.

- A. learning-based
- B. rules-based
- C. detective-based
- D. pattern-based

Correct Answer: B. rules-based

Explanation: Rules based systems (also called traditional or rule based) strictly follow a pre written set of "if then" commands and do not adapt over time. A simple calculator or basic alarm clock are examples. Learning based systems change with experience, and the other options are not standard terms.

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Q4 [Type: MCQ | Difficulty: Easy | QID:56694 ]

Fill in the blank: After the training process is finished, the AI 'brain' that is ready to work is called the \_\_\_\_\_.

- A. Algorithm
- B. Confuser
- C. Model
- D. Detective

Correct Answer: C. Model

Explanation: The model is the final software product that has successfully learned the patterns from the training data. Once training is complete, the model can be used to make predictions on new inputs. The algorithm is the method used during training, not the final brain.

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Q5 [Type: MCQ | Difficulty: Easy | QID:56695 ]

Fill in the blank: When an AI tells you its guess with a percentage, it is providing a \_\_\_\_\_.

- A. Rulebook
- B. Prediction
- C. Recap
- D. Teachable Machine

Correct Answer: B. Prediction

Explanation: A prediction is the AI's output or answer based on the patterns it learned during training. When a weather forecaster says there is an 80% chance of rain, that is a prediction. The percentage is often called a confidence score. Rulebook, recap, and Teachable Machine are not correct terms for this output.

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Q6 [Type: True/False | Difficulty: Easy | QID:56696 ]

True or False: A traffic light that changes every 30 seconds is an example of Machine Learning.

- A. False
- B. True

Correct Answer: A. False

Explanation: A standard traffic light follows a strict timer based rule and does not learn or change its behavior based on experience. It does not get better at managing traffic the more it watches cars. Therefore it is a rules based system, not machine learning.

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Q7 [Type: True/False | Difficulty: Easy | QID:56697 ]

True or False: Machine Learning systems can often improve and get better the more examples they see.

A. False

B. True

Correct Answer: B. True

Explanation: Increased exposure to data allows a machine learning system to refine its understanding of patterns and reduce errors. This is a key advantage of ML over traditional rules based systems. More examples generally lead to better performance.

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Q8 [Type: True/False | Difficulty: Easy | QID:56698 ]

True or False: If you show a trained AI something it has never seen before, it will always guess correctly.

A. False

B. True

Correct Answer: A. False

Explanation: AI can only recognize patterns that were present in its training data. Showing it something brand new, like a peace sign to an AI only trained on fists, often leads to confusion or incorrect guesses. AI does not have true understanding, only pattern matching.

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Q9 [Type: True/False | Difficulty: Easy | QID:56699 ]

True or False: Your brain naturally uses patterns to make predictions, such as knowing it might rain when you see dark clouds.

A. False

B. True

Correct Answer: B. True

Explanation: Human learning and AI learning both rely on recognizing repeating signals to predict future events. Your brain sees dark clouds (a pattern) and predicts rain. This similarity is why AI is often compared to how humans learn from experience.

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Q10 [Type: True/False | Difficulty: Easy | QID:56700 ]

True or False: In a rules-based system, the programmer is the one who figures out the rules.

A. False

B. True

Correct Answer: B. True

Explanation: In traditional (rules based) programming, humans must identify every condition and instruction for the computer to follow. The programmer figures out the rules and writes them explicitly. In machine learning, the computer figures out patterns from data instead.

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Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:56701 ]

Why is it difficult to use 'Rules' to catch every single spam email?

- A. Emails are always too long for rulebooks.
- B. Rules only work for math, not for words.
- C. Computers are not fast enough to read every email.
- D. Spammers are sneaky and constantly change how they write messages.

Correct Answer: D. Spammers are sneaky and constantly change how they write messages.

Explanation: Because rules are rigid, small changes like adding spaces, using synonyms, or misspelling words can easily bypass them. Spammers adapt quickly, so a fixed rulebook becomes outdated almost immediately. Machine learning is better at adapting to new spam patterns.

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Q12 [Type: MCQ | Difficulty: Easy | QID:56702 ]

Which of these apps uses Machine Learning to suggest videos based on what you have watched before?

- A. Alarm Clock
- B. YouTube
- C. A standard Calculator
- D. A Microwave

Correct Answer: B. YouTube

Explanation: YouTube uses recommendation engines that analyze your watch history and patterns to predict what videos you will enjoy next. The app gets smarter about your interests the more you use it. Alarm clocks, microwave buttons, and calculators do not use machine learning for recommendations.

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Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:56703 ]

What are the three simple steps AI takes to find and use patterns?

- A. Collect Examples, Find What's the Same, Predict New Examples
- B. Buy a Computer, Plug it in, Press Start
- C. Ask a Human, Type the Answer, Save the File
- D. Write Rules, Follow Instructions, Same Answer Every Time

Correct Answer: A. Collect Examples, Find What's the Same, Predict New Examples

Explanation: These three steps outline the core machine learning process: gathering data (collect examples), identifying patterns (find what's the same), and applying those patterns to new inputs (predict new examples). The other options describe traditional programming or unrelated actions.

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Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:56704 ]

If an AI shows a 'Confidence Score' of 95% for a 'Peace Sign', what does that mean?

- A. The AI has saved 95 photos so far.
- B. The AI is very sure that it is seeing a peace sign.
- C. The AI has finished 95% of its training.
- D. The AI is 95% broken and needs repair.

Correct Answer: B. The AI is very sure that it is seeing a peace sign.

Explanation: Confidence scores represent the mathematical probability the model assigns to its prediction. A higher percentage means the AI is more certain about its answer. 95% confidence means the AI strongly believes the input matches a peace sign pattern.

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Q15 [Type: MCQ | Difficulty: Easy | QID:56705 ]

Which part of the AI Recipe is the 'method for learning'?

- A. Model
- B. Prediction
- C. Algorithm
- D. Data

Correct Answer: C. Algorithm

Explanation: The algorithm is the mathematical procedure or recipe that allows the computer to process data and learn patterns. It is the method for learning. Data is the input, the model is the result, and prediction is the output.

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Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:56706 ]

In Google Teachable Machine, what is the computer doing when you click 'Train Model'?

- A. It is deleting all the photos to save space.
- B. It is asking a human to write down the rules for each hand gesture.
- C. It is scanning photos to find patterns and adjusting itself.
- D. It is searching the internet for the answer keys.

Correct Answer: C. It is scanning photos to find patterns and adjusting itself.

Explanation: Training is the phase where the algorithm analyzes the provided examples to create a functioning model. The computer looks for patterns in the photos and adjusts its internal parameters to learn the differences between classes (like peace sign vs. fist). It does not ask humans, delete photos, or search the internet.

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Q17 [Type: MCQ | Difficulty: Easy | QID:56707 ]

Which of these is the best example of a 'Pattern'?

- A. A sequence of numbers like 3,6,9,12...
- B. A random pile of leaves on the ground.
- C. A computer that is turned off.
- D. A one-time event that never happens again.

Correct Answer: A. A sequence of numbers like 3,6,9,12...

Explanation: A pattern is something that repeats or follows a predictable rule. The number sequence increases by 3 each time, which is a clear repeating relationship. Random piles, off computers, and one time events do not show repetition or predictability.

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Q18 [Type: MCQ | Difficulty: Easy-Medium | QID:56708 ]

How does AI use patterns to help you in Google Maps?

- A. It follows a rule that traffic is always bad on Mondays.
- B. It learns from millions of phones to predict traffic flow.
- C. It draws the map by hand using a digital pen.
- D. It calls your phone to ask if you are stuck in traffic.

Correct Answer: B. It learns from millions of phones to predict traffic flow.

Explanation: By analyzing the speed and location patterns of many users' phones, Google Maps' machine learning model can predict where traffic jams are likely to occur and estimate travel times. It does not use fixed rules, hand drawn maps, or phone calls.

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Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:56709 ]

What is the primary reason we need Machine Learning for Snapchat filters?

- A. To follow a rule that says 'all people look the same'.
- B. To make the phone battery last longer.
- C. To learn exactly where your eyes, nose, and mouth are.
- D. To record your voice and change the pitch.

Correct Answer: C. To learn exactly where your eyes, nose, and mouth are.

Explanation: Human faces vary wildly in shape, size, color, and angle. Machine learning is needed to recognize the "face pattern" across many different people and positions so that filters can stick to the correct facial features as you move. Simple rules cannot handle this variation.

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Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:56710 ]

In the context of the session, what does it mean to 'Think Like AI'?

- A. Looking for repeating details in examples to make a guess.
- B. Calculating math problems as fast as a calculator.
- C. Being able to see in the dark like a robot.
- D. Reading a rulebook and following it exactly.

Correct Answer: A. Looking for repeating details in examples to make a guess.

Explanation: Thinking like AI means identifying statistical regularities (patterns) in data and using those patterns to predict outcomes. It is about recognizing what repeats across examples, not about speed, night vision, or following fixed rulebooks.